

Anomalous aortic origin of a coronary artery in two children

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Abstract

A high proportion of the sudden deaths among young individuals is attributable to anomalous aortic origin of a coronary artery. Among congenital heart diseases, the incidence of congenital coronary artery anomaly is approximately 1.3%. Herein, we report 2 surgical cases of anomalous aortic origin of a coronary artery.

Keywords

Aorta, thoracic, cardiac surgical procedures, child, coronary vessel anomalies, heart arrest

Introduction

Anomalous aortic origin of a coronary artery (AAOCA) is a common cause of sudden death in young individuals. However, the incidence of coronary artery anomalies is approximately 1.3%.¹ In the past, this condition was often diagnosed incidentally during autopsy, but recently, detection of symptomatic and asymptomatic coronary artery abnormalities has been increasing owing to the development of imaging technology.

Case reports

Case 1

A 9-year-old boy had an episode of ventricular fibrillation that led to sudden loss of consciousness during marathon practice. Cardioversion was performed and adrenaline was administered; he recovered without neurological sequelae. After referral to our hospital, he was diagnosed with vasospastic angina pectoris, via cardiac catheterization. After discharge, he played a game of tag, lost consciousness, and had another episode of ventricular fibrillation. Again, he recovered without neurological sequelae. An electrocardiogram revealed negative T waves in leads V1–V2 but no significant ST changes (Figure 1). Transthoracic echocardiography showed the left coronary artery originating from the noncoronary sinus and an intramural segment from the vicinity of the commissure between the left

coronary cusp and noncoronary cusp (Figure 2). Coronary computed tomography (CT) revealed that the left coronary artery originated from the noncoronary sinus with an interarterial course but no stenosis of the left main trunk. Based on these findings, the cause of cardiopulmonary arrest was confirmed to be ischemia of the left coronary artery due to AAOCA. Surgery was performed under cardiopulmonary bypass. The left coronary artery was seen to originate from the noncoronary sinus with an intramural course to the left coronary sinus (Figure 3). The commissure between the left coronary cusp and noncoronary cusp was detached from the aortic wall. Unroofing of the intramural segment was performed with suturing of the incised intima of the aorta to the aortic wall. The detached commissure was repaired at a position slightly lower than normal; aortic valve insufficiency was not observed. The patient had no problem weaning from extracorporeal circulation. He was discharged on the 21st postoperative day. Currently, there has been no recurrence of arrhythmia.

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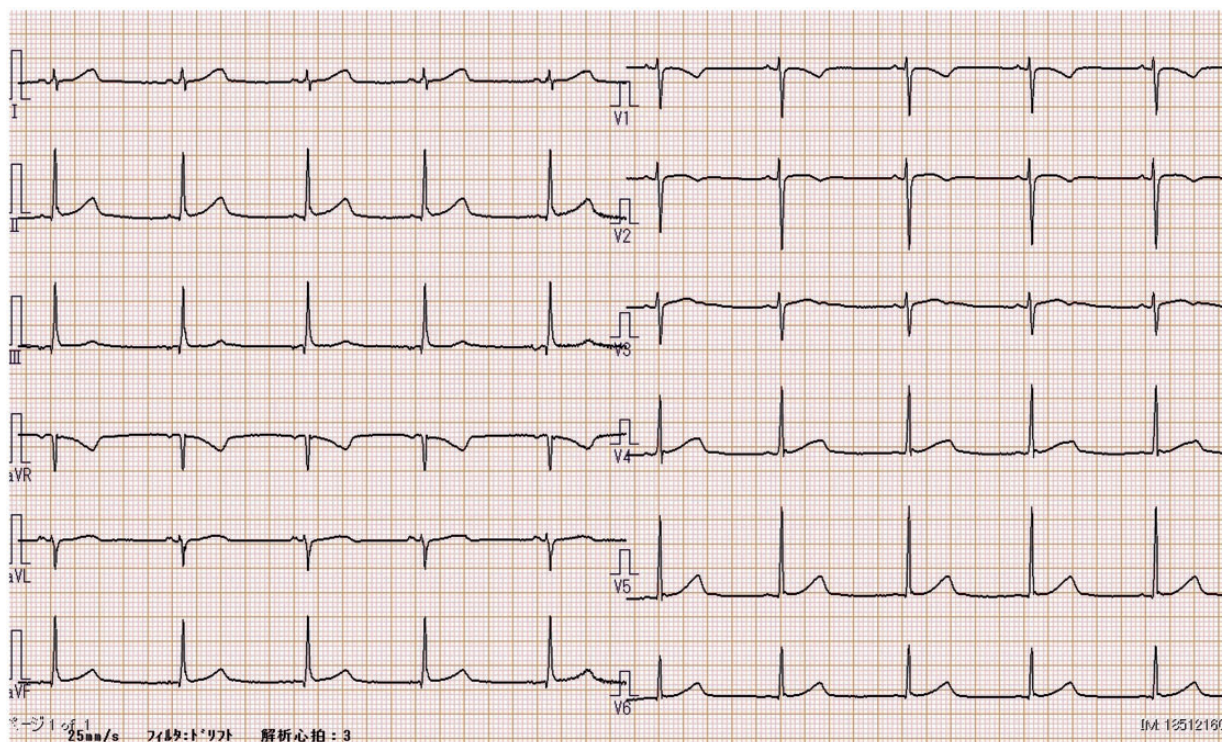


Figure 1. Electrocardiogram in case 1 showed negative T waves in leads VI-V2 with no significant ST changes or abnormal Q waves.

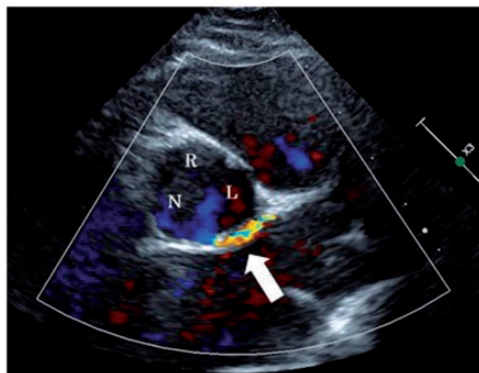


Figure 2. Preoperative transthoracic echocardiogram in case 1, revealing that the left anterior descending artery originated from the noncoronary sinus with an intramural segment (arrow).



Figure 3. Intraoperative view in case 1, showing that the orifice of the left coronary artery (arrow) originated from the non-coronary sinus near the commissure between the left coronary cusp and the noncoronary cusp.

Case 2

A 13-year-old girl suffered cardiopulmonary arrest while playing soccer. She recovered with higher brain dysfunction and leg paresis. No significant ST changes or abnormal Q waves were found on electrocardiography (Figure 4). Transthoracic echocardiography revealed that the left coronary artery originated from the right coronary sinus with an intramural segment (Figure 5). Coronary CT showed the left coronary artery originating from the right coronary sinus with

an interarterial course but no stenosis of the left main trunk. The patient was diagnosed with AAOCA, and surgery was performed under cardiopulmonary bypass. The left coronary artery was observed to originate from the right coronary sinus, and the left main trunk from the intramural segment (Figure 6). Unroofing of the intramural segment was carried out, similar to case 1. Aortic regurgitation was trivial, and there was no problem weaning from extracorporeal circulation. The patient was discharged on the 16th postoperative day. Currently, there has been no recurrence of arrhythmia.

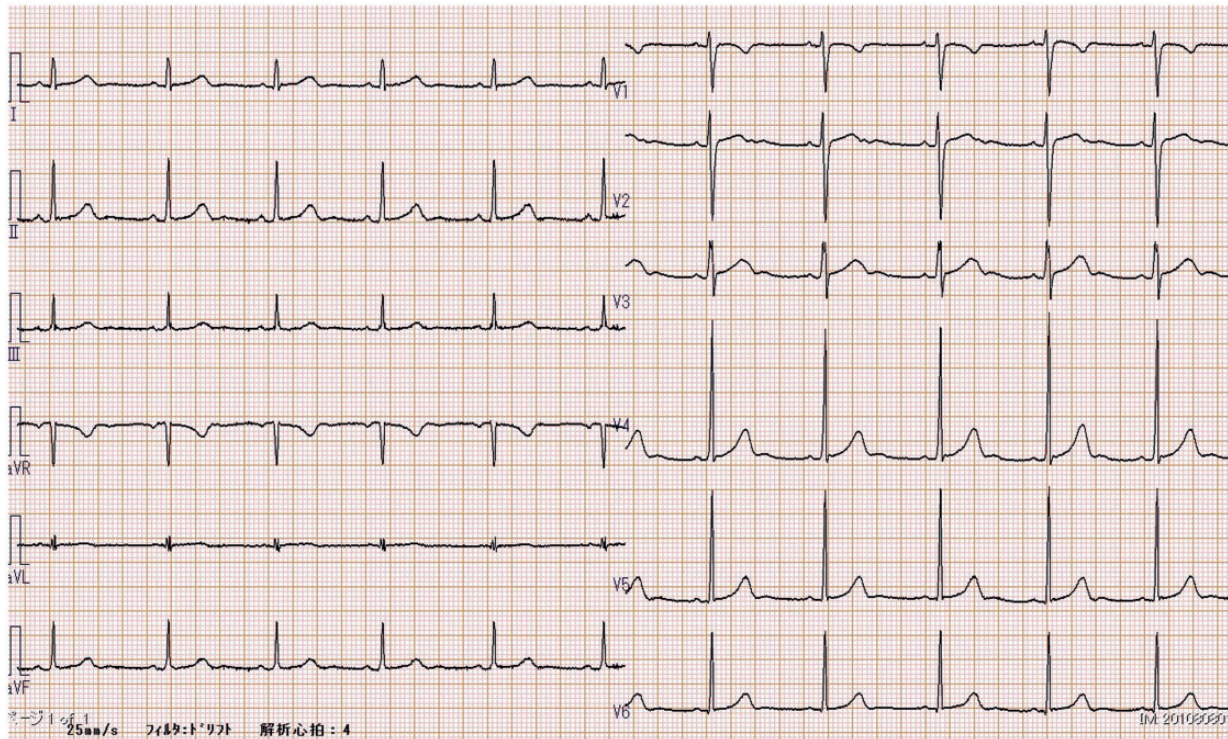


Figure 4. Electrocardiogram in case 2, showing no significant ST changes or abnormal Q waves.

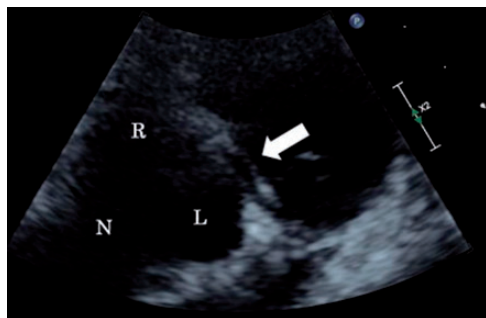


Figure 5. Preoperative transthoracic echocardiogram demonstrating the left anterior descending artery originating from the right coronary sinus with an intramural segment (arrow).

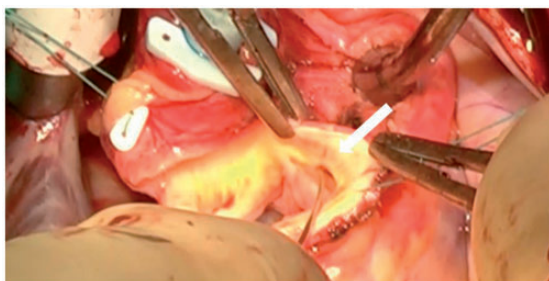


Figure 6. Intraoperative view showing that the orifice of the left coronary artery (arrow) originated from the right coronary sinus near the commissure between the left and right coronary cusps.

Discussion

Congenital coronary artery anomaly is a rare condition, and it has several variations. Sirasapalli and colleagues² claimed that among the anomalies of the coronary artery, 0.35% of AAOCA cases originate from the noncoronary sinus. As in case 1, the left coronary artery originating from the noncoronary sinus is extremely rare.²⁻⁴ The problem with the patient in case 1 was that he was diagnosed with vasospastic angina and lost consciousness again after discharge. Diagnosis of AAOCA via cardiac catheterization alone is challenging. The origin of the coronary artery must be assessed in more detail via coronary CT and transthoracic echocardiography. Diagnosis of an intramural course via coronary CT is often challenging in children. However, we believe that a more accurate diagnosis can be made when a patient is assessed via both echocardiography and coronary artery CT.

The unroofing method is useful in surgery for AAOCA. Mery and colleagues³ recognized recurrent aborted sudden cardiac death after unroofing of a short intramural segment. Although the exact mechanism is unknown, they suggested that translocation and neo-ostium creation should be considered in cases of a short intramural segment. Our patients had long intramural segments, therefore, we believe that unroofing was suitable. However, both patients were children so it is necessary to follow up for stenosis of the new

origin of the coronary artery. The unroofing method is considered a basic method for AAOCA, although patients should be followed up closely for aortic regurgitation owing to the possibility of detachment of the commissure.⁵

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

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